

Data Mining Project Report



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Weather Recognition Project Report

1. Introduction

This project focuses on developing a weather recognition system using machine learning techniques to classify weather conditions from images. The system is designed to recognize various weather types such as cloudy, fog/smog, lightning, rain, rainbow, sandstorm, shine, snow, and sunrise.

2. Project Overview

The project utilizes a dataset of weather images categorized into nine different classes. The main steps include data loading, preprocessing, feature extraction using PCA (Principal Component Analysis), and classification using a Support Vector Machine (SVM) model.

3. Key Components:

- **Data Loading and Preprocessing:** Images are loaded, resized, and flattened for processing.
- **Feature Scaling:** Standard scaling is applied to normalize the data.
- **Dimensionality Reduction:** PCA is used to reduce the feature dimensions while retaining 95% of the variance.
- **Model Training and Evaluation:** An SVM model is trained and evaluated on the preprocessed data.

4. Methodology

- **Data Loading:** Loading images from the dataset
- **Loading Process:** Images are loaded using OpenCV, resized to 32×32 pixels, and flattened into 1D arrays
- **Data Preprocessing:**
 - **Train-Test Split:** The dataset is split into training (80%) and testing (20%) sets
 - **Standard Scaling:** Features are standardized to have zero mean and unit variance
 - **PCA:** Applied to reduce dimensionality while preserving 95% of the variance, improving computational efficiency

5. Model Selection

Support Vector Machine (SVM) is a powerful supervised machine learning algorithm widely used for classification tasks. It works by finding the optimal hyperplane that best separates data points of different classes in a high-dimensional space.

- **Margin Maximization:** SVM aims to maximize the margin between different class boundaries, ensuring better generalization.
- **Support Vectors:** Only the data points closest to the decision boundary (called support vectors) influence the position and orientation of the hyperplane.

SVM performs well even with limited training data and high-dimensional feature spaces, making it suitable for image classification tasks like weather recognition.

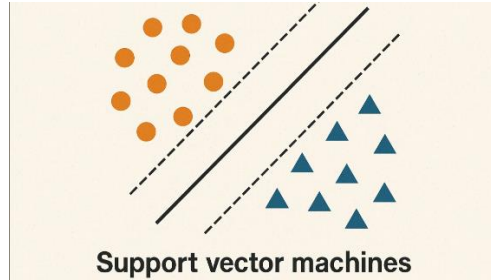


Figure 1: Model Selection

In this project, we use the **RBF (Radial Basis Function) kernel**, which is effective for non-linear classification. The model learns to find an optimal hyperplane that separates different classes in the transformed feature space. SVM is trained on the PCA-reduced feature set, and performance is evaluated.

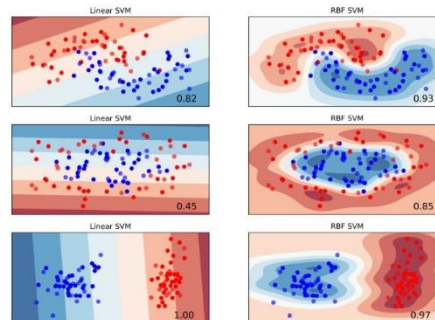


Figure 2: Kernel Selection

6. Evaluation

- Accuracy: 70.18%
- Confusion Matrix: Provides insights into the model's performance across different classes.

```
print("SVM Accuracy:", accuracy_svm)
print("SVM Confusion Matrix:\n", conf_matrix_svm)
print("SVM Macro F1 Score:", macro_f1_svm)
```

```
[4]:
SVM Accuracy: 0.7017543859649122
SVM Confusion Matrix:
[[35 10 11  7  4  0  7  0  0]
 [ 5 72  3  7  1  3  2  2  0]
 [ 4  2 52  4  0  0  1  0  0]
 [ 1  5  5 62  1  2  0 10  1]
 [ 5  8  6  3 17  2  5  0  1]
 [ 1  9  0  3  1 68  0  0  4]
 [ 1  1  2  0  0  0 39  3  0]
 [ 0  9  0 14  2  1 144  0]
 [ 0  0  1  1  0  4  1  0 59]]
SVM Macro F1 Score: 0.6925569942890968
```

Figure 3: Model Evaluation

7. Prediction Example

- Testing Image

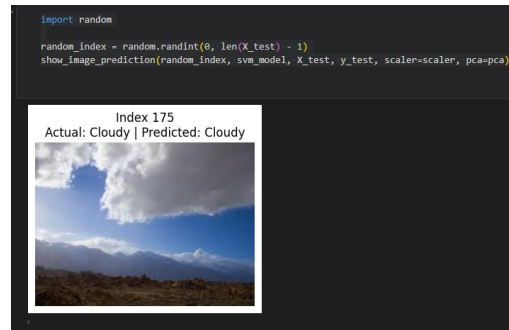


Figure 4: Model prediction output for a weather image (Testing Dataset)

- External Image



Figure 5: Model prediction output for a weather image (External Image)

8. Challenges and Improvements

- **Data Imbalance:** Some classes (e.g., rainbow) have fewer images, which may affect model performance. Techniques like data augmentation or class weighting could be explored.
- **Feature Extraction:** PCA helps reduce dimensionality.
- **Model Selection:** Experimenting with other classifiers (e.g., Random Forest, Knn) or tuning SVM hyperparameters could improve results.
- **Image Quality:** Enhancing image quality or using more diverse datasets might help the model generalize better.

9. Conclusion

The weather recognition system demonstrates promising results with an accuracy of 70.18%. Future work could focus on improving the model by addressing data imbalance, exploring advanced feature extraction techniques, and experimenting with different classifiers. This project serves as a foundation for more sophisticated weather recognition systems, which could be valuable for applications like automated weather monitoring and forecasting.